

SEAN PEREIRA

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Summary

Innovative full-stack developer and ML engineer with strong research foundation in AI/ML optimization, scalable system architecture, and algorithm development. Passionate about contributing to cutting-edge research and emerging technologies. Proven ability to design and implement production-grade applications with focus on performance optimization and innovative problem-solving.

Education

Fr. Conceicao Rodrigues College of Engineering (CRCE), Mumbai Bachelor of Engineering in Computer Engineering CGPA: 8.5 / 10.0	<i>Expected May 2027</i>
Sardar Vallabhbhai Patel Junior College, Mumbai 12th Grade (HSC) Grade: 71.17%	2023
St. Francis (ICSE) School, Mumbai 10th Grade (ICSE) Grade: 97.5%	2021

Experience

Technical Trainee – E-Cell, IIT Guwahati - Completed intensive training in backend architecture, scalable web system design, AI/ML applications, and startup ecosystem methodologies. - Helped businesses improve digital presence, increasing user engagement by 50%+ and contributing to 15% revenue growth.	<i>February – March 2024</i>
Documentation Head – Codelabs CRCE - Organized college coding contests and technical seminars and collaborated with design and backend teams for seamless execution. - Placement preparation sessions for 150+ third-year students and streamlined documentation and event workflows. - Designed session content with industry mentors to connect technical experts and students.	<i>March 2024 – March 2025</i>

Projects

EduSphere – Immersive Collaborative Learning Platform <i>Technologies: React.js, Three.js, Node.js, Express.js, MongoDB, Socket.io, Tailwind CSS, WebGL</i> - Architected a full-stack virtual learning hub with multi-user synchronization, real-time message broadcasting, and persistent community data channels using Socket.io and Express.js. - Integrated customized 3D assets and dynamic interactive viewport environments directly into the student landing system using Three.js and WebGL. - Implemented fluid asset optimization strategies including script/model lazy loading and viewport rendering throttling, preserving a consistent 60+ FPS responsive UI across heterogeneous devices.
CancerCare AI – RAG-Powered Oncology Support Platform <i>Technologies: React.js, Python, FastAPI, RAG, FAISS, PostgreSQL, Llama-3.3, NLP</i> - Engineered a clinical NLP platform using Retrieval-Augmented Generation (RAG) for grounded, hallucination-resistant oncology support responses. - Designed an automated processing pipeline for unstructured oncology literature, building optimal text-chunking strategies and generating dense vector embeddings for low-latency indexed searches inside FAISS. - Implemented LLM-as-a-Judge evaluation framework with rigorous safety gating.

Technical Skills

Programming Languages:	Python, JavaScript, Java, C, SQL
Frontend Technologies:	React.js, Three.js, HTML, CSS, JavaScript, Tailwind CSS
AI / Machine Learning:	TensorFlow, Scikit-learn, NLP (NLTK, spaCy), RAG
Backend Frameworks:	FastAPI, Flask, Node.js, Express.js, Socket.io
Databases:	PostgreSQL, MongoDB, MySQL
Tools & Platforms:	Git/GitHub, Docker, AWS, Google Cloud, Jupyter, Colab, Figma, PowerBI, Vercel, Linux
Soft Skills:	Problem-Solving, Team Collaboration, Technical Communication, Leadership, Adaptability

Certifications & Achievements

- Google Cloud Engineer – Google Skills
- AWS Cloud Practitioner – AWS Academy
- Neural Network for NLP and CV – (Elite + Gold) By IIT Guwahati (NPTEL)
- AI for Management – (Elite + Silver) By IIT Madras (NPTEL)
- Data Analytics with Python – (Elite) By IIT Roorkee (NPTEL)
- Intellectual Property Rights and Competition Law – (Elite) By IIT Kharagpur (NPTEL)
- Cleared **GATE CSE 2026**
- 3rd Place** – Hackathon 3.0, IES College of Management & Research, Bandra (2025)